# Statistical Supervised Machine Learning

## TP 5 – PAC-Bayes for majority votes: certify, then learn

## Master MIASHS & MALIA

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**Abstract**

Last of five practical sessions (1: protocol, 2: bagging, 3: boosting, 4 and 5: PAC-Bayes). This session follows Sections 15 to 17 of the lecture notes. Bagging, random forests and boosting all output a majority vote; TP 4 certified the Gibbs classifier, which draws one voter at random, and we now certify the *vote* itself. We first relate the two with the first-order bound and see that it ignores how the errors of the voters are correlated; the joint error and the disagreement of pairs of voters then give the tandem bound, which we study on voters whose correlation we control. We then certify the random forest of the running example of the notes *without any test set*, with its out-of-bag examples, and we finally *learn* the weights of the forest by minimizing a bound: we derive the self-bounding algorithm of Section 17 step by step, implement it, and see why the bound to minimize must take the diversity of the voters into account. The session lasts **three hours**; Part 5 contains the difficult questions.

**About this session – TP 5: PAC-Bayes for majority votes**

**Goal.** By the end of the session you should be able to compute the Gibbs risk, the joint error and the disagreement of a vote, prove and compute the first-order and tandem bounds, certify a random forest with its out-of-bag examples, derive a self-bounding algorithm from a bound, implement it, and explain why minimizing the first-order bound destroys the diversity of a vote.

**To reread before the session.** TP 2 (random forests, Part 4) and TP 4, and in the lecture notes: Section 15 (from the Gibbs classifier to the majority vote: Definition 15.1, Propositions 15.2, 15.3, 15.5 and 15.6), Section 16 (learning with the bounds, out-of-bag examples) and Section 17 (deriving a self-bounding algorithm, Propositions 17.1 and 17.2, Algorithm 13). The exercises of Sections 15 to 17 of the exercise booklet are a good preparation.

**Timing (3 hours).**

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| Part 1 | 25 min |
| Part 2 | 45 min |
| Part 3 | 45 min |
| Part 4 | 65 min |
| Part 5 (⋆) | at home |

**To hand in.** Your notebook with the tables of Parts 1 to 4, the figures of Parts 2 and 4, and your proofs.
**Tools.** `numpy`, `scipy`, `matplotlib`, `DecisionTreeClassifier`; the toolbox `common/pbtools.py` provides the bounds of the course (`fo_bound`, `tnd_bound`) and Algorithm 13 (`tandem_fixed_point`) to check your own functions.

**Notation (Section 15).** $\mathcal{H}$ is a finite set of voters $h : \mathcal{X} \to \{-1, +1\}$ and $Q$ a distribution on $\mathcal{H}$. For an example $(x, y)$, $W_Q(x, y) = \sum_h Q(h)\mathbb{1}[h(x) \neq y]$ is the $Q$-weighted proportion of voters that err (Definition 15.1). We use

- the **Gibbs risk** $R_{\mathcal{D}}(G_Q) = \mathbb{E}[W_Q] = \sum_h Q(h) R_{\mathcal{D}}(h)$ and the risk of the **majority vote** $R_{\mathcal{D}}(B_Q) = \mathbb{P}(B_Q(X) \neq Y)$, where $B_Q(x) = \operatorname{sign}\left(\sum_h Q(h)h(x)\right)$ (ties counted as errors);
- the **joint error** $e_{\mathcal{D}}(Q) = \sum_{h,h'} Q(h)Q(h')\,\mathbb{P}(h(X) \neq Y, h'(X) \neq Y)$ and the **disagreement** $d_{\mathcal{D}}(Q) = \sum_{h,h'} Q(h)Q(h')\,\mathbb{P}(h(X) \neq h'(X))$ (Definition 15.4).

The empirical versions are obtained by replacing $\mathcal{D}$ by a sample. As in TP 4, $\overline{\mathrm{kl}}^{-1}(q, \varepsilon)$ is the upper inverse of the binary kl, and $\delta = 0.05$.

> **Good practice** Store the predictions of the voters on a sample in a matrix `V` of shape (voters, examples) and the errors in `E = (V != y)`. Then the empirical risks are `E.mean(1)`, the matrix of empirical joint errors is `E @ E.T / n`, the Gibbs risk of $Q$ is `Q @ L` and its joint error `Q @ Lt @ Q`.

## 1 Gibbs classifier, majority vote and the first-order bound (Sections 15.1–15.2)

1. Show that $B_Q$ errs on $(x, y)$ if and only if $W_Q(x, y) \geq \frac{1}{2}$, and deduce from Markov's inequality (Lemma 13.1) that $R_{\mathcal{D}}(B_Q) \leq 2R_{\mathcal{D}}(G_Q)$ (Proposition 15.2).
   Combine it with Seeger's bound (TP 4) to obtain a certificate for the vote.
2. On the toy problem of TP 4 (one sample, $m = 100$, uniform prior on the 102 thresholds), compute for the Gibbs posteriors $Q_\lambda$, $\lambda \in \{10, 30, 100, 300, 3000\}$: the true Gibbs risk, the true risk of the vote (on a fine grid of $[0, 1]$, since it can be computed exactly), the oracle bound $2R_{\mathcal{D}}(G_Q)$ and the PAC-Bayesian first-order bound.
   Comment.

## 2 Condorcet, joint error, disagreement and the tandem bound (Sections 15.3–15.5)

We consider $n = 25$ voters with uniform weights and the following model: voter $i$ errs on an example when $\sqrt{c}\,Z + \sqrt{1-c}\,\xi_i < \Phi^{-1}(p)$, where $Z$ (the "difficulty" of the example) and $\xi_1, \dots, \xi_n$ are independent $\mathcal{N}(0,1)$ and $\Phi$ is the c.d.f. of $\mathcal{N}(0,1)$. Every voter has risk $p$, and $c \in [0,1)$ tunes the correlation between their errors ($c = 0$: independent errors). Given $Z$, the errors are independent Bernoulli variables of parameter $\pi(Z) = \Phi\big((\Phi^{-1}(p) - \sqrt{c}Z)/\sqrt{1-c}\big)$, so all the exact quantities below can be computed by a one-dimensional integral over $Z$.

1. **Condorcet.** For $c = 0$, prove Proposition 15.3: $R_{\mathcal{D}}(B_Q) \le e^{-2n(\frac{1}{2}-p)^2}$.
   Compute the exact risk of the vote for $n \in \{1, 5, 25, 101, 1001\}$, $p \in \{0.3, 0.45\}$, $c \in \{0, 0.3\}$, compare it with this bound when $c = 0$, and give its limit when $n \to \infty$.
   Conclude.
2. **Identity.** Show that $R_{\mathcal{D}}(G_Q) = e_{\mathcal{D}}(Q) + \frac{1}{2} d_{\mathcal{D}}(Q)$ (Proposition 15.5; hint: for two voters, $\mathbb{1}[h \neq y] + \mathbb{1}[h' \neq y] = 2\,\mathbb{1}[h \neq y,\, h' \neq y] + \mathbb{1}[h \neq h']$), and that $e_{\mathcal{D}}(Q) = \mathbb{E}[W_Q^2]$.
3. **Tandem bound.** Show that $R_{\mathcal{D}}(B_Q) \le \mathbb{P}(W_Q^2 \ge \frac{1}{4}) \le 4 e_{\mathcal{D}}(Q)$ (Proposition 15.6).
4. Simulate the errors of the 25 voters on $m = 20\,000$ examples for $p = 0.3$, $c = 0.3$. Compute the Gibbs risk, the joint error and the disagreement (a) from $W_Q$, (b) from the pairs of voters (all the $n^2$ ordered pairs, including $h = h'$), and the risk of the vote.
   Compare with the exact values.
5. For $p \in \{0.2, 0.35, 0.45\}$ and $c \in [0, 0.98]$, plot the risk of the vote and the two oracle bounds: first-order $2R_{\mathcal{D}}(G_Q)$ and tandem $4e_{\mathcal{D}}(Q)$.
   When is the tandem bound better than the first-order bound? What does it say about diversity?

## 3 Certifying a random forest without a test set (Sections 15.5 and 16.2)

We use the running example of Part IV of the notes: `digits`, task "is the digit smaller than 5?", split with `train_test_split(X, y, test_size=0.3, random_state=0)` (1257 training and 540 test digits), and a forest of 100 trees, each learned with `DecisionTreeClassifier(max_features="sqrt")` on a bootstrap sample of *half* the size of the training set. The prior $P$ is uniform on the *indices* $1, \dots, 100$ of the trees, fixed before the forest is grown. Each tree $h_t$ is evaluated on its *out-of-bag* (OOB) examples $T_t$, the training examples that are not in its bootstrap sample.

1. Write `grow_forest(Xtr, ytr, n_trees, frac, rng)` returning the trees and the boolean $100 \times m$ matrix of the OOB examples.
   Show that an example is out-of-bag for a given tree with probability $(1 - \frac{1}{m})^{m/2} \approx e^{-1/2} \approx 61\%$, and for two given trees with probability $\approx e^{-1} \approx 37\%$. Check it.

2. Compute the OOB error $\hat{L}(h_t)$ of each tree on $T_t$, and the empirical joint error $\hat{e}(h_t, h_{t'})$ of each pair on $T_t \cap T_{t'}$; let $n_1 = \min_t |T_t|$ and $m = \min_{t,t'} |T_t \cap T_{t'}|$.
For the uniform posterior $Q = P$, compute the first-order certificate $2\overline{\mathrm{kl}}^{-1}\left(\sum_t Q(h_t)\hat{L}(h_t), \frac{\mathrm{KL}(Q\|P)+\ln(2\sqrt{n_1}/\delta)}{n_1}\right)$ and the tandem certificate $4\overline{\mathrm{kl}}^{-1}\left(\hat{e}(Q), \frac{2\,\mathrm{KL}(Q\|P)+\ln(2\sqrt{m}/\delta)}{m}\right)$ (equations (15.3) and (15.4), with the OOB replacements of Section 16.2).

3. Compute on the test set the risk of the vote, the Gibbs risk, the joint error and the disagreement, and the kl test-set bound of the vote (Theorem 13.5).
Compare the three certificates. Which ones need a test set?

4. Why are the OOB certificates valid, although every training example has been used to grow some of the trees?
Why must $m$ be the size of the *smallest* common OOB set?

## 4 Learning the weights of the forest: deriving a self-bounding algorithm (Sections 16.3 and 17)

Since the bounds hold for all posteriors $Q$ simultaneously, we may *choose* the weights of the 100 trees by minimizing a bound on the OOB data, and keep the value of the bound as a certificate (self-bounding learning, Section 16.1). We follow the five steps of Section 17.1.

1. **The first-order recipe.** With the bound of Theorem 14.4 applied to the OOB errors, the objective is $\sum_t Q(h_t)\hat{L}(h_t) + \frac{\mathrm{KL}(Q\|P)}{\lambda}$, minimized by the Gibbs posterior $Q_\lambda(h_t) \propto P(h_t)e^{-\lambda\hat{L}(h_t)}$ (Proposition 14.6).
For $\lambda \in \{1, 2, 4, \ldots, 4096\}$, compute its first-order certificate, its number of *effective trees* $1/\sum_t Q(h_t)^2$ and the test risk of its vote.
Comment.

2. **The tandem objective.** Applying Theorem 14.4 to pairs of voters gives the linear tandem bound (Proposition 17.1). Show that, for a fixed $\lambda$, its terms that depend on $Q$ are $F_\lambda(Q) = \hat{e}(Q) + \frac{2}{\lambda}\,\mathrm{KL}(Q\|P)$.

3. **The optimality condition.** With a Lagrange multiplier for the constraint $\sum_t Q(h_t) = 1$ (Section 4.3), show that a minimizer $Q^\star$ of $F_\lambda$ with positive weights satisfies $Q^\star(h) \propto P(h)\, e^{-\lambda\hat{e}(h,Q^\star)}$, with $\hat{e}(h, Q) = \sum_{h'} Q(h')\hat{e}(h, h')$ (Proposition 17.2).
Interpret: how is a tree penalized, compared with the Gibbs posterior?

4. **The algorithm.** Implement the damped fixed-point iterations of Algorithm 13 ($\alpha = 0.1$) and check your function against `tandem_fixed_point` of `common/pbtools.py`.
For $\lambda \in \{1, 2, 4, \ldots, 4096\}$, compute the tandem certificate of the output, its number of effective trees and the test risk of its vote.
Plot the certificates and the test risks of the two algorithms against $\lambda$, and the weights of the trees against their OOB errors for the selected $\lambda$.

5. Why may $\lambda$ be chosen by minimizing the kl certificate, without any union bound? Is the certificate still valid if the iterations are stopped before convergence?

6. Apply the two algorithms (with $\lambda$ selected on the certificate) to `spambase`, `wdbc` and `australian` (split 70%/30%, `random_state=0`, same forests).
   Report, for the uniform weights and for the two learned posteriors, the certificate, the number of effective trees and the test risk of the vote.
   Which bound is the right objective for learning a vote?

## 5 Going further ($\star$)

1. **The factor 2 cannot be improved.** Take $n$ voters ($n$ odd) with uniform weights, a set $A$ of inputs of probability $\frac{2rn}{n+1}$ on which exactly $\frac{n+1}{2}$ voters err, and no error outside $A$.
   Compute $R_{\mathcal{D}}(G_Q)$, $e_{\mathcal{D}}(Q)$, $R_{\mathcal{D}}(B_Q)$, the ratio $R_{\mathcal{D}}(B_Q)/R_{\mathcal{D}}(G_Q)$ and the tandem bound for $n \in \{3, 11, 101\}$, $r \in \{0.05, 0.2\}$.
   Conclude.

2. **The C-bound.** Using the Chebyshev–Cantelli inequality $\mathbb{P}(Z - \mathbb{E}Z \geq a) \leq \frac{\mathrm{Var}Z}{\mathrm{Var}Z + a^2}$ for the margin $M_Q = 1 - 2W_Q$ (Section 15.6), prove the C-bound $R_{\mathcal{D}}(B_Q) \leq 1 - \frac{(1-2R_{\mathcal{D}}(G_Q))^2}{1-2d_{\mathcal{D}}(Q)}$ when $R_{\mathcal{D}}(G_Q) < \frac{1}{2}$. Add it to the figure of Part 2, question 5.

3. **Convexity.** When all the voters are evaluated on the same examples $x_1, \ldots, x_m$, show that $\hat{e}(Q) = \frac{1}{m}\sum_i W_Q(x_i)^2$, and deduce that $F_\lambda$ is convex, so that the fixed point of Algorithm 13 is its global minimum.

4. **The linear certificate.** For the posterior selected in Part 4, compute the linear tandem certificate of Proposition 17.1 with the union bound over the 13 values of $\lambda$ ($\ln \frac{13}{\delta}$ instead of $\ln \frac{1}{\delta}$) and compare it with the kl certificate. Why is it looser?